Quan (Kyle) Zhang

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Education

Ph.D., Economics (Quantitative Economics & Econometrics), University of Hawai'i at Mānoa 2023 – 2027
M.A., Economics (en route to Ph.D.), University of Hawai'i at Mānoa 2026
M.S., Quantitative Health and Clinical Research, John A. Burns School of Medicine, University of Hawai'i at Mānoa 2024 – 2026

Research Interests

Health and labor economics (health shocks, work, and retirement); economics of aging and the family, including frailty dynamics over the life course; causal inference and applied econometrics; quantitative analytic methods for longitudinal cohort data with complex survey designs (weighting, stratification, replicate-weights variance estimation); health policy, health equity, and disparities; quantitative health sciences and clinical research methodology, including cost-effectiveness analysis, psychometric measurement, and biomarker-integrated longitudinal study design; disaster recovery and community resilience.

Peer-Reviewed Publications

1. Yuan, K., Ye, Y., Zhou, X., Gu, Y., Jiang, Y., Yin, S., **Zhang, Q.**, Yang, H., Sang, X., Jiang, L., Zhuang, Z., Xu, K., & Huang, S. Integrated multi-omics and machine learning analyses to improve and explore molecular subtypes and immune profiles of primary hepatocellular carcinoma. *Discover Oncology*, 2026. DOI
2. Zhang, Q., & Fu, Y. Ensuring trustworthy AI assisted guideline development for clinical practice. *npj Digital Medicine*, 2026, 9(1), 661. DOI
3. Zhang, Q. Do negative social ties accelerate aging in adults, or does aging erode social ties? *Proceedings of the National Academy of Sciences (PNAS)*, 2026, 123(20), e2608036123. DOI
4. Zhang, Q. Keep REMS focused on safety, not as a barrier to generics. *Health Affairs*, 2026, 45(4), 451. DOI
5. Zhang, Q. Enhancing life-course analyses of financial disadvantage and depressive mood through longitudinal design and causal inference. *Journal of Epidemiology*, 2026, 36(5), 173–174. DOI
6. Zhang, Q. Overcoming bias in a fall prevention program in resource-limited areas. *JAMA*, 2026, 335(4), 370–371. DOI
7. Zhang, Q. Operationalizing distributional cost-effectiveness in genomic medicine: A United States–focused comment. *Value in Health*, 2026, 29(1), 172–173. DOI
8. Zhang, Q. On "The joint impact of greenspace and air pollution on mortality": Methodological proposals. *International Journal of Epidemiology*, 2026, 55(1), dyaf224. DOI
9. Zhang, Q. Re: Khodakarami et al, "Exploring neighborhood opportunity as a factor in pediatric asthma visits to the emergency department": A causal roadmap for Child Opportunity Index 2.0. *Public Health Reports*, 2025. DOI

GBD 2023 Collaborative Publications (Group Authorship, Under Review)

1. GBD 2023 Colon and Rectum Cancer Collaborators. "Global, Regional, and National Burden of Colon and Rectum Cancer from 1990 to 2023, with Forecasts to 2050: A Systematic Analysis for the Global Burden of Disease Study 2023." Under review, *The Lancet Oncology*. (IHME ID: 4219; named collaborator-author.)
2. GBD 2023 Western Pacific WHO AMR Collaborators. "The Burden of Bacterial Antimicrobial Resistance in the WHO Region of the Western Pacific: A Cross-Country Systematic Analysis." Under review, *The Lancet Regional Health – Western Pacific*. (IHME ID: 4006; named collaborator-author.)
3. GBD 2023 Type 1 Diabetes Collaborators. "Global, Regional, and National Type 1 Diabetes Burden: A Systematic Analysis of the Global Burden of Disease Study 2023." Under review, *The Lancet Primary Care*. (IHME ID: 5338; named collaborator-author.)
4. GBD 2023 Alzheimer's Collaborators. "Global, Regional, and National Burden of Alzheimer's Disease and Other Dementias in 204 Countries and Territories, 1990–2023, with Forecasts to 2050: A Systematic Analysis for the Global Burden of Disease Study 2023." Under review, *The Lancet*. (IHME ID: 6217; named collaborator-author.)

Working Papers

1. Zhang, Q. "Working Through Frailty: Chronic versus Acute Health Shocks and the Margins of Labor Supply." SSRN
2. Zhang, Q., & Hu, W. "Subclinical Cardiometabolic–Inflammatory Burden, Disability-Free Working Life and Work Exit in Three Ageing Cohorts."
3. Zhang, Q., Lyu, J., Slavens, C., & Hu, W. "Sustained Insulin Resistance and Incident Coronary Heart Disease in Older English Adults: A Selection-Weighted Cohort Analysis of the English Longitudinal Study of Ageing." SSRN
4. Zhang, Q. "Health Shocks and the Permanent Labor-Market Exit of Older Europeans."
5. "Nutritional Depletion, Inflammation, and Colorectal Cancer in US Adults, 2001–2023" (with Z. Wang and Y. Fu).

Invited Contributions

1. Zhang, Q. "Social Determinants of Health and Multidimensional Poverty." In R. Baikady (Ed.), *The Palgrave Encyclopedia of Concepts and Theories in the Context of Global Poverty*. Palgrave Macmillan / Springer Nature, 2026. DOI
2. Zhang, Q. "Multidimensional Poverty Index (MPI): Measurement, Health Deprivations, and Policy Use." In R. Baikady (Ed.), *The Palgrave Encyclopedia of Global Poverty Research*. Palgrave Macmillan / Springer Nature. Under review.

Conference Presentations & Workshops

1. Zhang, Q. "Working Through Frailty: Chronic versus Acute Health Shocks and the Margins of Labor Supply." Southern Economic Association (SEA) 96th Annual Meeting, session "Health and Work," Houston, TX, November 21, 2026.
2. Zhang, Q. "Health Insurance Design and Work Exit After Chronic Disease Onset: Harmonized Evidence from the United States, England, and China." Gateway Research Workshop on Cross-Country Comparisons (Gateway to Global Aging Data), held during the Gerontological Society of America (GSA) 2026 Annual Scientific Meeting, National Harbor, MD, November 4, 2026.
3. AWARD Network Summer Training Institute (selected participant, one of 22 early-career scholars), University of California, San Francisco, San Francisco, CA, July 20–24, 2026.

Research Experience

Graduate Research Assistant — UHERO, University of Hawai'i at Mānoa 2024 – Present

Maui Wildfire Exposure Study (MauiWES) *2024 – Present*

NIH- and state-funded longitudinal cohort of 2,000+ participants, in collaboration with UHERO, JABSOM, and UH Maui College.

- Supported data collection: data entry, on-site support at data-collection events, and descriptive analyses (distributions and summary statistics); acknowledged contributor to the team's 2025 *JAMA Network Open* study of health and social support after the Maui wildfires.

Kaiona Survey Project *2025 – Present*

Youth health and well-being survey and program evaluation, in partnership with Wai'anae Coast Comprehensive Health Center.

- Built Qualtrics consent, demographic, survey, and intake instruments; cleaned and managed survey data; produced pre–post data reports for three youth summer programs.

Roots Reborn Program Evaluation *2025 – 2026*

Post-wildfire economic empowerment intervention.

- Designed and executed full evaluation of an IRB-approved human-subjects intervention: Qualtrics survey design → matched-panel construction → pre–post causal analysis → executive summary for policymakers.
- Built session-level evaluation framework linking program costs to quality-adjusted participant outcomes, producing a submission-ready manuscript with actionable policy recommendations.

Small Business Success in Honolulu Study *2024 – 2025*

City & County of Honolulu Office of Economic Revitalization. Co-author of the UHERO report *Investigating and Tackling Barriers and Needs for Small Business Success in Honolulu* (2025).

Research Allied Health Professional — The Queen's Medical Center, Honolulu 2025 – Present

- Approved for data collection and analysis on an IRB-approved colorectal cancer study drawing on electronic health records and tumor specimens, in collaboration with the Department of Quantitative Health Sciences, JABSOM.

GBD Senior Collaborator — Global Burden of Disease 2023, IHME, University of Washington 2025 – Present

- Provided substantive peer review and contributions for nine GBD 2023 estimation manuscripts spanning cancer, cardiovascular and neurodegenerative disease, metabolic disease, dietary and environmental risks, antimicrobial resistance, and population ageing.
- Contributions recognized with named collaborator-authorship on four GBD 2023 papers currently under review at *The Lancet*, *The Lancet Oncology*, *The Lancet Regional Health – Western Pacific*, and *The Lancet Primary Care*.
- Promoted to **GBD Senior Collaborator** (September 2026), the tier granted for co-authorship on four or more GBD publications, carrying eligibility to lead GBD papers as first author and to join scientific consultations on methods and models.

Teaching Experience

Instructor, ECON 321 Introduction to Statistics, Department of Economics, UH Mānoa Summer 2025
Teaching Assistant, Principles of Microeconomics (ECON 130) & Macroeconomics (ECON 131) 2023 – 2024

Awards & Honors

2026	Research Stipend, Department of Economics, University of Hawai'i at Mānoa
2026	AWARD Network Summer Training Institute, University of California, San Francisco (NIA/NIH-funded)
2025, 2026	Burnham Campbell Fellowship Award, Department of Economics, University of Hawai'i at Mānoa
2024 – 2026	Hung Family Fellowship, University of Hawai'i at Mānoa (AY 2024–25 and 2025–26)
2023	Achievement Award, Department of Economics, University of Hawai'i at Mānoa

Professional Service

Journal Peer Reviewer: *Public Health in Practice* (Elsevier), 2025–Present · *Aging* (Aging-US), 2026
Conference Abstract Reviewer: Gerontological Society of America (GSA) 2026 Annual Scientific Meeting, 2026
University Service: Reviewer, Graduate Student Organization Merit-Based Awards (Empowerment & Leadership category), University of Hawai'i at Mānoa, 2026
Senior Collaborator & Peer Reviewer, Global Burden of Disease Study 2023, Institute for Health Metrics and Evaluation (IHME), University of Washington, 2025–Present (Senior Collaborator since September 2026). Manuscripts reviewed (IHME ID): Global Type 1 Diabetes (5338)†; Antimicrobial Resistance in the WHO Western Pacific Region (4006)†; Global Colon & Rectum Cancer (4219)†; Global Alzheimer's Disease & Other Dementias (6217)†; Global Environmental Exposures (6176); Global Dietary Risks (6115); Global Ageing and Non-Communicable Diseases (5410); Global Stroke (6155); Global Ageing and Inequality (5409). †Named collaborator-author.

Professional Memberships

American Economic Association (AEA) · American Society of Health Economists (ASHEcon) · International Health Economics Association (iHEA) · Society for Epidemiologic Research (SER) · Gerontological Society of America (GSA) · Hawaii Pacific Gerontological Society (HPGS) · Hawaii Economic Association (HEA) · Pi Gamma Mu

Technical Skills

Statistical Software: R (tidyverse, fixest, modelsummary, survey, srvyr, ggplot2, Shiny), Stata 18 (including `svy` suite for complex-survey estimation), Python (pandas, statsmodels, scikit-learn), SQL, Git.
Complex Survey & Longitudinal Methods: Stratified multistage cluster designs, longitudinal and cross-sectional weighting, replicate-weights variance estimation (BRR, jackknife, Taylor linearization), wave-to-wave attrition adjustment, nonresponse weighting.
Causal Inference & Econometrics: Two-way fixed effects, difference-in-differences, instrumental variables (2SLS), event study designs, regression discontinuity, propensity score matching, inverse probability weighting, Heckman selection models, mediation analysis, sensitivity analysis (Oster 2019; Cinelli–Hazlett 2020), Lee bounds, specification curve analysis, wild cluster bootstrap, power analysis.
Machine Learning & Computational Methods: Double/debiased machine learning (DML), causal forests, LASSO/elastic net, multiple imputation (MICE).
Quantitative Health Sciences: Frailty index construction (Rockwood deficit accumulation), item response theory (IRT), confirmatory factor analysis (CFA), GAD-7/CES-D psychometric scoring, cost-effectiveness analysis (CEA/DCEA), survey design (Qualtrics).
Survey & Data Platforms: HRS, ELSA, SHARE, CHARLS, NHANES, MEPS, BRFSS, IPUMS (USA/CPS/International), ATUS, GBD.

Research Training & Certifications

CITI Program: Responsible Conduct of Research; Biomedical Research Investigators and Key Personnel; Non-Exempt Biomedical Researchers and Key Personnel (incl. IPS); Good Clinical Practice (2025–2026) · IHME Introduction to GBD (2025)

Languages

English (fluent) · Mandarin Chinese (native)